Sabitri Panda

Date of Birth: 13th Aug 2004
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Summary

Final-year BS–MS student in Physics with strong theoretical training in quantum mechanics and statistical physics, and hands-on analytical and numerical research experience in condensed matter and quantum many-body systems. Research interests include disordered superconductivity, topological phases, Floquet and non-equilibrium dynamics, quantum spin liquids, and phase-space methods in quantum information. Seeking a PhD position in condensed matter physics.

Education

Indian Institute of Science Education and Research (IISER), Kolkata — Mohanpur, West Bengal
BS–MS in Physics | *CGPA :* 8.4/10.0 — *2022 – 2027*

Army Public School, R. K. Puram — Secunderabad, Telangana
Class 12th *(CBSE)* | *Boards (Overall):* 93.8% — *2022*

Army Public School, R. K. Puram — Secunderabad, Telangana
Class 10th *(CBSE)* | *Boards (Overall):* 97.2% — *2020*

Research Experience

1.Long Range Coulumbic Repulsion in Disordered Superconductivity. — 2026 – Present
Master's Thesis

Mentor: Prof. Amit Ghosal | ***IISER Kolkata***

I am investigating how long-range Coulomb repulsion affects superconductivity in a disordered two-dimensional periodic lattice. In clean superconductors, screening allows the repulsion to be absorbed into a pseudopotential μ^*; with disorder, electron localization weakens screening, so the long-range interaction becomes a crucial ingredient. Within a Bogoliubov–de Gennes mean-field framework, I aim to study how the repulsion modifies the superconducting pairing amplitude, its spatial inhomogeneity, and screening. I am currently implementing Ewald summation numerically to treat the slowly decaying $1/r$ interaction under periodic boundary conditions, splitting it into rapidly converging real-space and reciprocal-space sums.

2. Sub-Planckian structures & quantum geometry of nonclassical states. — 2025–Present
Project

Mentor: Prof. Sourin Das | ***IISER Kolkata***

I studied coherent-state formalism (Weyl–Heisenberg, SU(2), SU(1,1)) and the Wigner-function description of cat and compass states, deriving the interference structure of compass states and the emergence of sub-Planckian features underlying enhanced displacement sensitivity I extended this to N-component cat states, analyzing their Wigner and Husimi-Q functions and dependence on N and amplitude. Among quantum-information measures, Shannon and Wehrl entropies could not resolve the fine structure, whereas quantum Fisher information, fidelity, and Heisenberg scaling captured the quantum-metrological advantage. I derived the relation between the quantum metric and the quantum Fisher information, giving a geometric interpretation of the sensing precision. Building on this interpretation, I am studying the Aharonov–Anandan geometric phase and a time-dependent geodesic Hamiltonian generating evolution along geodesics of the quantum-state manifold.

3. Periodically driven quantum spin liquid. — 2025 – 2026
Project

Mentor: Prof. Vishwanath Shukla | ***IIT Kharagpur***

I studied the Kitaev honeycomb model and its fractionalization into Majorana fermions and gauge sectors, solving it through both the Jordan–Wigner transformation and Majorana fermionization, followed by diagonalization. I extended the Majorana formulation to the Floquet unitary operator of time-periodic drives, including protocols with n drives, and derived the energy (quasi-energy) spectrum analytically under periodic boundary conditions(2D periodic sectors). I also investigated prethermalization and thermalization in driven spin systems using high-frequency expansions, examining their implications for fractionalized dynamics.

4. Quantum transport in disordered tungsten thin films (PPMS). 2025
Project

Mentor: Dr. Partha Mitra | ***IISER Kolkata***

We measured the resistivity of three ∼20 nm tungsten films (α, β, and mixed phase) as a function of temperature (down to 0.34 K, using a He-3 probe) and magnetic field in a PPMS, using the van der Pauw method. I analyzed the low-temperature conductivity in terms of weak localization and variable-range hopping, and interpreted the magnetoresistance as negative at low fields (suppression of weak localization), positive at intermediate fields (two-band model), and saturating at high fields, with a crossover to classical behavior near 3 K.

5. Topological phases in 1D quantum lattice models (SSH model). Summer 2025
Internship

Mentor: Prof. Vishwanath Shukla | ***IIT Kharagpur***

I studied the Su–Schrieffer–Heeger chain by deriving its Bloch Hamiltonians (periodic and canonical), band structure, and density of states, and verifying its inversion, time-reversal, and chiral symmetries. I numerically diagonalized finite open chains (20 and 50 sites) to track zero-energy edge states for $v < w$ and their finite-size splitting near the transition, and derived the edge-state profile and decay length analytically. I computed the winding number, Zak phase (π versus 0), and polarization ($e/2$) to establish the bulk–edge correspondence.

6. Thermal phenomena and Brownian motion. Summer 2024
Internship

Mentor: Prof. Arindam Kundagrami | ***IISER Kolkata***

I studied the transport of momentum, heat, and particles starting from the continuity equation and Fick's laws, and derived the viscosity, thermal conductivity, and self-diffusion coefficients of a dilute gas using kinetic theory and the mean free path, including their pressure and temperature dependence and the limits of the relation $\kappa = c_V \eta$. I then examined Brownian motion through Einstein's diffusion approach and the Langevin equation, linking them via the Stokes–Einstein relation, and reviewed its role in statistical mechanics and in colloidal, biological, and financial applications.

Technical Skills

Programming Languages: Python, Julia, HTML

Libraries: NumPy, SciPy, QuSpin, Pandas. Matplotlib

Numerical and theoretical Methods: Exact diagonalization, Bogoliubov–de Gennes mean-field theory, Floquet theory, Wigner function formalism

Scientific Writing: LaTeX (Overleaf)

Scholarship

Merit cum Means Scholarship McM
Awarded by Indian Institute of Science Education and Research, Kolkata *2025-Present*

Co-Curricular Activities

Gold in Stage Play (Drama) IICM 2025

Silver in Photography IICM 2025

Ex-Office Bearer, IISER Kolkata Drama Club 2023-2024

References

Prof. Amit Ghosal Master's Thesis Advisor
Professor, IISER Kolkata ***ghosal@iiserkol.ac.in***

Prof. Sourin Das Project Guide
Professor, IISER Kolkata ***sourin@iiserkol.ac.in***

Prof. Vishwanath Shukla Internship Guide
Professor, IIT Kharagpur ***vishwanath.shukla@phy.iitkgp.ac.in***